

Project Title: Nine Lives

Overview

Nine Lives is a command-line roguelike survival game. The player starts with 9 lives and advances across an endless glass bridge by stepping on one glass panel at each step. Each stage consists of multiple steps, and each step contains several panels. Exactly one panel is safe, and the others break. The player tries to survive as long as possible by choosing paths and using reward items. The game ends when the player's lives become 0.

Term Definition

- Panel: a selectable glass tile in a step.
- Step: one event where the player either uses an item or chooses a panel.
- Stage: a group of multiple steps.
- Sector: a group of 5 stages.
- Path: a difficulty route that determines the number of panels in each step and the reward tier after the sector.

Requirements

1. The game runs in the terminal as a command-line game, and the player starts at Stage 1 with 9 lives.
2. There is no maximum life limit, so healing items can increase the player's lives above the starting value.
3. Each stage consists of multiple steps. The number of steps per stage starts at 2 and increases by 1 every 10 stages. For example, Stages 1–10 have 2 steps per stage, Stages 11–20 have 3 steps per stage, and Stages 21–30 have 4 steps per stage.
4. A sector consists of 5 stages. At the start of each sector, the player chooses one path by entering a number: 1 for Safety, 2 for Normal, and 3 for Abyss.
5. The chosen path determines both the number of panels in each step and the reward

- tier after the sector: Safety has 2 panels and gives a Silver reward, Normal has 3 panels and gives a Gold reward, and Abyss has 4 panels and gives a Prism reward.
6. In each step, exactly one panel is randomly selected as the safe panel, and all other panels are breaking panels.
 7. Panels are numbered from left to right. In Safety, the available panel numbers are 1 and 2. In Normal, the available panel numbers are 1, 2, and 3. In Abyss, the available panel numbers are 1, 2, 3, and 4.
 8. At the beginning of each step, the game shows the player's inventory. The player may enter 0 to use no item, or enter an item number to use the corresponding item. The player can use at most one item in each step.
 9. If an item is used, the selected item takes effect immediately and is removed from the inventory. After the item-use phase, the player chooses a panel.
 10. If the player chooses a panel, the player enters the panel number. If the chosen panel is safe, the player proceeds to the next step without losing a life. If the chosen panel is breaking, the player loses 1 life and proceeds to the next step.
 11. If the player's lives become 0, the game ends immediately.
 12. After every 5 completed stages, the game shows the two items in the reward tier of the completed path, and the player chooses one item to add to the inventory.
 13. Silver rewards contain First Aid Kit and Rubber Hammer. First Aid Kit gives +2 lives, and Rubber Hammer removes one breaking panel from the current step before panel selection.
 14. Gold rewards contain Large First Aid Kit and Reinforced Shoes. Large First Aid Kit gives +5 lives, and Reinforced Shoes prevent life loss from breaking panels until the current stage ends.
 15. Prism rewards contain Full Recovery and Prism Shield. Full Recovery gives +9 lives, and Prism Shield takes effect immediately and prevents life loss from breaking panels until the end of the next stage.
 16. For every user input, if the player enters a non-number input or a number outside the valid range for the current prompt, the game prints an error message and asks the player to enter the input again without changing the game state.
 17. When the game ends, the game prints the final stage reached, total cleared steps, path history, and number of used items.

Example interaction

The game starts at Stage 1 with 9 lives. The user chooses a path by entering 1 for Safety, 2 for Normal, or 3 for Abyss. If the user chooses Safety, each step has two panels numbered from left to right. At the beginning of a step, the game shows the user's inventory and asks whether to use an item. The user enters 0 to use no item, then selects a panel number. If the selected panel is safe, the user proceeds to the next step without losing a life. If the selected panel breaks, the user loses 1 life and proceeds to the next step. After the user completes 5 stages, the game gives a reward based on the chosen path. When the user's lives become 0, the game ends and prints the final stage reached, total cleared steps, path history, and number of used items.